Bryan Nguyen

714-710-2342 bryann3528@gmail.com LinkedIn Portfolio

Education

University California Irvine **Expected Graduation: 2027**
Pursuing Bachelor of Science in Mechanical Engineering - GPA: 3.5

Skills

Mechanical & CAD: SOLIDWORKS, OnShape, AutoCAD, 3D Printing
Robotics & Embedded: Arduino, Raspberry Pi, Sensor Integration, Motor Control, Isaac Lab
Programming Languages: MATLAB, C++, Python

Projects

Robotic Manipulation System Integration (LeRobot) **Jan 2026 - Present**

- Configured teleoperation using the **LeRobot** framework to control a follower arm and collect demonstration data
- Trained a task-specific control policy from teleoperated demonstrations to perform object grasping and placement
- Deployed and evaluated the learned policy on hardware, successfully executing block pick-and-place into a target bin

Ball Balancing Robot **Dec 2025 - Jan 2026**

- Built a 3-DOF delta robot which balances a ball on a platform, using **SOLIDWORKS** and **3D-printed** components
- Implemented real-time ball tracking in **Python** using **OpenCV**, processing camera frames to compute ball position error relative to target setpoints
- Developed inverse kinematics and **PID control** algorithms in **Python** to convert vision-based position error into joint angle commands for three servo motors

5-DOF Robotic Arm **Jul 2025 - Sept 2025**

- Designed, prototyped, fabricated and tested a 5-DOF robotic arm, with a payload capacity of **250g**
- Designed full mechanical assembly in **SOLIDWORKS** and **3D printed** structural parts and custom gears using ABS, PLA and PA6-GF
- Built custom electronics and power system utilizing LM2596 buck regulation, NEMA-17 base rotation, servos for joints, hand wired and soldered
- Embedded firmware in **C++** on an **Arduino Nano** to read four potentiometers from a master controller, driving servos and stepper motor for real-time teleoperation

Solid Rocket Motor Development **Jan 2025 – Jun 2025**

- Led a 5-person engineering team in the design, machining, and static testing of a solid rocket motor, achieving **2000 N** peak thrust and **5776 Ns** total impulse
- Designed propulsion hardware in **OnShape** with emphasis on manufacturability, structural integrity, and assembly for ground testing
- Executed static fire tests to collect thrust and burn rate data, then analyzed results and modeled system performance using **MATLAB**
- Integrated the propulsion system into a dual-deployment vehicle and validated performance through flight testing, reaching altitudes over **7000 ft**

Experience

Solid Rocket Fuel Research **Sept 2024 – May 2025**
Cal Poly Pomona ENGAGE Program

- Conducted experimental static testing of propulsion materials, collecting burn rate and thrust data under controlled conditions
- Analyzed test results to evaluate system performance and support design decisions using **MATLAB**

Rocket Workshop **Sept 2024 – Jan 2025**
Workshop Lead

- Designed and co-lead a 10-week workshop for **30 students**, covering CAD, simulation, and fabrication of rockets
- Organized meetings, budgets, and logistics, adapting to challenges like relocating fabrication space
- Provided hands-on mentorship, guiding students through design challenges and technical skills